

Pavle Slepčev

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EDUCATION

University of Pittsburgh, Swanson School of Engineering
B.S. Mechanical Engineering, Minor in Space Engineering

Expected: May 2028

Relevant Schoolwork: Simulation and Design | Dynamic Systems | Circuits and Systems | Mechanical Design I | Space Engineering I

ENGINEERING EXPERIENCE

Drivetrain Sub-Member – Panther Racing (Formula SAE)

May 2026 – Present

- Researched and rated gear load capacity in KISSsoft across ISO 6336, DIN 3990, and AGMA methods, selecting tooth-thickness and center-distance tolerances that control backlash within machinable limits.
- Authored an onboarding deck on the 2027 FSAE rules to train new members on the rulebook and tech inspection.
- Assisted with redesigning a two-stage planetary gearbox in KISSsoft.

Aerodynamics Sub-Team Member – Panther Racing (Formula SAE)

August 2025 – May 2026

- Designed a rear wing swan neck from simulated and hand-calculated aerodynamic loads, organizing load cases in Excel and iterating the geometry in ANSYS Mechanical until the design met its force requirements.
- Collaborated with the Composites Lead to design and finish front, side, and rear wing airfoil molds.
- Laid up carbon fiber and epoxy resin to produce M-plates and body panels.

PROJECTS

WASP — GNSS-Denied Drone Navigation

Jan 2026 – Present

- Co-developed a quadcopter navigation system that fixes absolute position by registering live camera frames against pre-downloaded georeferenced satellite imagery, removing the drift error of visual and inertial odometry.
- Designed a 3D-printed carbon-fiber nylon (PAHT-CF) mount to carry an NVIDIA Jetson Orin Nano and IMX477 camera on a modified DJI Phantom 3. Performed incremental dead-load flight testing to obtain a payload ceiling.
- Sized a six-damper isolation system to an 18 Hz natural frequency, and verified the mount in SOLIDWORKS at a 4× gravity landing case.
- Flight-tested over seven flights producing 13 camera-to-satellite position fixes; the final two averaged 8 m horizontal error against onboard GNSS specified to ±3 m.

Custom Mechanical Keyboard PCB

- Designed a keyboard PCB end to end in KiCad — schematic capture, component selection, footprint assignment, and board layout — carrying the design through a full production flow rather than assembling a kit.

Idi-Run — iOS Application

- Built a SwiftUI application on a Supabase backend that turns GPS run traces into a territory-capture game, pairing run tracking (distance, pace, history) with map-based multiplayer and leaderboards.

6-Axis Robotic Arm (BCN3D Moveo)

- Open-sourced a bill of materials to design and build a 6 DOF robotic arm with a payload capacity of a full kilogram. 3D-printed structure driven by five steppers and a servo from an Arduino Mega with RAMPS 1.4 and TB6560 drivers.

Bridge Project

Formula SAE

- In MATLAB, I created an optimization force analysis script to find the best truss design for force and cost based on predetermined material prices. The code outputted pin positions, angles, total forces, and minimum material usage, which allowed us to calculate costs.

ADDITIONAL EXPERIENCE

Basketball Coach, Jewish Community Center — Pittsburgh, PA

2023 – 2024

Carpentry and Demolition, Melnyk Contracting

2024-26

SKILLS

Design & Analysis: SOLIDWORKS, ANSYS Mechanical, KISSsoft, AutoCAD, Fusion 360, KiCad, OpenRocket

Programming: MATLAB, Simulink, Python, C/C++, Swift/SwiftUI, JavaScript

Manufacturing: 3D printing, carbon fiber layup, laser cutting, CNC, hand and power tools

Community Engagement

International Food Channel: I host weekly episodes of cooking international recipes

2026

Club Volleyball Officer, University of Pittsburgh

2026

ACHIEVEMENTS & AWARDS

- **Statics 0135 Bridge Competition** — 2nd place overall (performance index 190) and 3rd in total weight held (1000lbf).
- **Seal of Biliteracy** | Serbian
- **All-City Pittsburgh Basketball League MVP**
- **National Volleyball Scholar and Player**
- **CSWP | Certified SolidWorks Professional**